

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE

CONTINUOUS ASSESSMENT TEST 1 - JANUARY-2026

CET05103 - BASIC VISUAL PROGRAMMING

SEMESTER: III

YEAR: 2

DURATION: 1 Hr 30 Min.

MAX. MARKS: 50

CURATED ANSWERS PROVIDED BY 4LAZIE - SMART IN BRAIN

PART A: SHORT ANSWER QUESTIONS

Q1. What does the acronym BASIC stand for?

Answer: BASIC stands for Beginner's All-purpose Symbolic Instruction Code.

Smart Tip: Remember "Beginner's" as it was originally designed for students to easily learn programming.

Q2. What is meant by "Visual" in Visual Basic?

Answer: "Visual" refers to the method used to create the Graphical User Interface (GUI). Instead of writing lines of code to build an interface, you can simply drag and drop pre-built objects (like buttons and textboxes) onto a screen.

Q3. What is a Module in Visual Basic?

Answer: A Module is a file (with a .vb extension) that contains code such as variables, constants, and procedures which can be accessed from anywhere within the project. It helps in organizing code and making it reusable.

Q4. List two numeric data types supported in Visual Basic.

Answer:

- **Integer:** Used for whole numbers (e.g., 10, -5).
- **Double:** Used for numbers with decimals (e.g., 3.14, -0.5).

Q5. What is the purpose of "Exit For"?

Answer: "Exit For" is a statement used to immediately terminate a For or For Each loop before it has finished all its iterations. It transfers execution to the statement immediately following the loop.

Q6. State the keyword used to declare a variable in Visual Basic.

Answer: The keyword 'Dim' (short for Dimension) is used to declare a variable in Visual Basic.

Q7. What is the purpose of Console.WriteLine()?

Answer: Console.WriteLine() is a built-in method used to display output text or data on the console screen, followed by moving the cursor to a new line.

PART B: EXPLANATORY QUESTIONS

Q8. Explain why visual basic has slower execution speed?

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Answer: Visual Basic typically has a slower execution speed because it is an interpreted, managed language (compiled to Intermediate Language, then interpreted by the runtime) rather than being compiled directly to native machine code. Its heavy reliance on GUI components and runtime libraries also adds overhead.

Q9. Explain the difference between method and function.

Answer:

- **Function:** A block of code that performs a specific task and ALWAYS returns a value to the calling code.
- **Method (Sub procedure):** A block of code that performs a specific task but DOES NOT return a value. In VB, these are called 'Sub' procedures.

Q10. Explain how to declare and initialize an array in visual basic with examples.

Answer: An array is declared using the 'Dim' keyword, followed by the variable name, parentheses () to indicate the size, and the 'As' keyword for the data type.

```
' Declaration  
Dim numbers(4) As Integer
```

```
' Initialization  
numbers(0) = 10  
numbers(1) = 20
```

Smart Tip: You can also declare and initialize on one line: `Dim numbers() As Integer = {10, 20, 30}`

Q11. Discuss the limitations of Visual Basic.

Answer:

- **Platform Dependency:** It is primarily designed to run on Windows operating systems.
- **Slower Execution:** As a managed language, it is slower compared to low-level languages like C or C++.
- **Memory Usage:** It consumes more memory due to the overhead of the .NET Framework and garbage collection.

Q12. Provide a short history of Visual Basic.

Answer: Visual Basic was created by Microsoft and first released in 1991 as a modern, event-driven successor to the older BASIC language. It revolutionized programming by introducing a drag-and-drop graphical interface builder. In 2002, Microsoft released Visual Basic .NET (VB.NET), integrating it fully into the .NET framework and making it a true object-oriented language.

Q13. Write a simple Visual Basic program to enter and display user name and age.

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Answer: Here is a simple console application to achieve this:

```
Module Module1
    Sub Main()
        Console.Write("Enter your name: ")
        Dim name As String = Console.ReadLine()

        Console.Write("Enter your age: ")
        Dim age As Integer = Convert.ToInt32(Console.ReadLine())

        Console.WriteLine("Hello " & name & ", you are " & age & " years old.")
        Console.ReadLine()
    End Sub
End Module
```

PART C: EXPLANATORY AND PROGRAMMING QUESTIONS

Q14. Analyze the advantages and disadvantages of using the Visual Basic IDE for system-level versus application-level programming.

Answer:

- **Application-Level Programming:** Excellent for this because the IDE provides rapid application development (RAD) tools, easy drag-and-drop GUI creation, and rich built-in libraries. However, users must have the .NET framework installed.

- **System-Level Programming:** High-level abstractions make code readable, but it is not suitable for system-level programming (like writing drivers or OS kernels) because it lacks direct, low-level memory access (like pointers in C) and has higher execution overhead.

Q15. Discuss the various Decision-Making structures available in Visual Basic.

Answer:

- **If...Then:** Executes a block of code only if a specified condition is true.
- **If...Then...Else:** Executes one block if true, and a different block if false.
- **If...ElseIf...Else:** Tests multiple conditions sequentially.
- **Select Case:** Tests a single expression against a list of matching values. It is much cleaner than multiple ElseIf statements when checking the same variable against many possible values.

Q16. Analyze the different types of Loops in Visual Basic with simple example program.

Answer:

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There are three primary types of loops in Visual Basic:

- **For...Next Loop:** Runs a block of code a specific, pre-determined number of times.
- **While...End While Loop:** Runs continuously as long as a specified condition remains true.
- **Do...Loop:** Can test the condition at the start (Do While) or at the end (Do Until), ensuring the code runs at least once if checked at the end.

Example of a For Loop:

```
Module Module1
    Sub Main()
        For i As Integer = 1 To 5
            Console.WriteLine("Number: " & i)
        Next
        Console.ReadLine()
    End Sub
End Module
```

Q17. Explain in details about multi dimension array and the looping process with an example program.

Answer: A multi-dimensional array is an array that contains more than one dimension, representing data in a grid or table format (like rows and columns). A 2D array is the most common. To loop through a multi-dimensional array, nested loops are used. The outer loop typically goes through the rows, while the inner loop iterates through the columns.

```
Module Module1
    Sub Main()
        ' Declare and initialize a 2x2 multi-dimensional array
        Dim matrix(,) As Integer = {{1, 2}, {3, 4}}

        ' Loop through rows
        For row As Integer = 0 To 1
            ' Loop through columns
            For col As Integer = 0 To 1
                Console.Write(matrix(row, col) & " ")
            Next
            Console.WriteLine() ' New line after each row
        Next
        Console.ReadLine()
    End Sub
End Module
```